



INTERDEPARTMENTAL LETTER OF ARRANGEMENT

FOR

COLLABORATIVE RESEARCH PROJECTS

IN SUPPORT OF

MATERIALS ACCELERATION DEMONSTRATION ECOSYSTEM ("MADE")
(hereinafter referred to as "Arrangement")

BETWEEN: **HIS MAJESTY THE KING IN RIGHT OF CANADA** as represented by the **NATIONAL RESEARCH COUNCIL OF CANADA**, established pursuant to the National Council Research Act R.S.C., 1985, N-15, having its headquarters located at 1200 Montreal Road, Ottawa, Ontario K1A 0R6 (the "**NRC**")

AND: **HIS MAJESTY THE KING IN RIGHT OF CANADA**, as represented by the **Minister of Natural Resources**, established pursuant to the Department of Natural Resources Act S.C., 1994, c. 41, having its headquarters located at 580 Booth Street, Ottawa, Ontario K1A 0E4 ("**NRCan**")

(Collectively referred to as the "**Participants**" or individually as a "**Participant**".)

WHEREAS under the Laboratories Canada Initiative, the Participants are working to establish a world-class collaborative research hub built around new innovative artificial intelligence (AI) driven, computer-enabled robotics technology to accelerate discovery and innovation of new materials technologies (the "**TerraCanada Mississauga/Hamilton Centres of Expertise**").

WHEREAS, within the TerraCanada Mississauga/Hamilton Centres of Expertise, the Participants intend to enter into an interdepartmental memorandum of understanding to develop a Materials Acceleration Demonstration Ecosystem ("**MADE**") to integrate the materials science and engineering capabilities of both organizations and develop and deploy a set of integrated science and technology platforms, with a central focus on Materials Acceleration Platforms ("**MAPs**") technology to accelerate materials discovery and development of new technologies and materials across a range of clean energy, structural (including additive manufacturing) and smart materials applications.

WHEREAS it is expected MADE will utilize facilities, both existing and in development, of both Participants, which are located in NRCan-CanmetMATERIALS Hamilton and NRC-Mississauga advanced manufacturing research facilities.

WHEREAS, as defined by the Science Statement of Requirements ("**SSoR**") dated 5 November 2019 and attached hereto as Annex B, the Participants agree that MADE's research mandate should include Accelerated Materials Discovery and Development Process; Materials and Process Sustainability; and Safety, Production Scale-up, Demonstration and Standardization (the "**Areas of Interest**").

WHEREAS the Participants share the objective of supporting MADE and wish to leverage the unique strengths of each organization by working collaboratively on research projects in the Areas of Interest, while ensuring alignment with each Participant's respective departmental mandates (the "**Purpose**").

IN CONSIDERATION, the Participants have come to the following understanding:



1. DEFINITIONS

In this Arrangement, unless the context requires otherwise, the following expressions shall have the following meanings:

- 1.1 **“Arising IP”** means Intellectual Property that is first conceived, or developed in a Joint Project and that is disclosed in the Deliverables.
- 1.2 **“Background IP”** means any Intellectual Property, other than Arising IP, developed outside a Joint Project and disclosed by one Participant to the other for the purpose of a Joint Project.
- 1.3 **“Deliverables”** means the results of a Joint Project, such as reports, physical models, samples, data records, drawings, and machine-readable software, which are specifically mentioned in the Statement of Work and Deliverables attached to a JPA as being deliverable.
- 1.4 **“Government Purposes”** means any use by a Participant in support of its mandate where such use does not result in the disclosure of any protected or classified information of the other Participant nor the licensing of the other Participant’s IP to any third parties.
- 1.5 **“Intellectual Property”** or **“IP”** means inventions (whether patentable or not); works or creative expression; computer programs; compilations of literary works; architectural works; artistic works; audio and video recordings; logos and other visual identity marks; 3D models; bacterial, viral, plant, human, or animal material that has new genetic or other characteristics; discoveries, records of experiment, scientific formulae, data, analyses, models, prototypes, specifications, algorithms, products, research tools, devices, know-how and related Confidential Information or trade secrets; as well as any similar intangible assets recognized from time to time in any jurisdiction. It also includes exclusionary rights such as: patents; copyrights; trademarks and official marks; industrial designs and design patents; integrated circuit topographies and mask works; plant breeders’ rights; and applications for any of the above, as well as any similar exclusionary rights recognized from time to time in any jurisdiction with respect to some types of Intellectual Property.
- 1.6 **“Joint Project”** means a collaborative research project undertaken jointly by the Participants in support of the Purpose which is detailed and defined in a JPA. Joint Project shall not include any project involving a third party.
- 1.7 **“Joint Project Arrangement”** or **“JPA”** means an arrangement substantially in the format provided in Annex A attached hereto which details the Joint Project and shall include each Participants contributions and a mutually agreed upon SOWD.
- 1.8 **“SOWD”** means the agreed to statement of work and deliverables attached to each JPA in the form identified in Schedule 1 of Annex A.
- 1.9 **“Jointly Created Arising IP”** means Arising IP that is jointly developed by both Participants in a Joint Project.
- 1.10 **“License to Occupy”** means the arrangement between the NRC, through its Real Property Planning and Management branch, and NRCan that manages the terms and conditions of the NRCan’s access and use, including placement of its equipment, of the Licensed Premises, as well as its general access to the NRC Building.
- 1.11 **“Licensed Premises”** means the areas being used by NRCan within the NRC building, to be specified in the License to Occupy.
- 1.12 **“NRC Building”** means the NRC’s Advanced Materials Research Facility at 2620 Speakman Dr, Mississauga, Ontario.



1.13 “**Solely Created Arising IP**” means Arising IP that is solely created by only one Participant in a Joint Project.

1.14 “**Tangible Property**” means any samples, equipment, vehicles, or materials that the one Participant provides to the other Participant to enable that Participant to perform its portion of a Joint Project, as described in a JPA’s SOWD.

2. SCOPE

2.1 The scope of activities under this Arrangement shall be in support of the Purpose. Specifically:

- (a) collaboration on research projects in the Areas of Interest and in other research areas that are subsequently identified as an approved focus of MADE; and
- (b) exchange of scientific and technical information relating to the development of work on Joint Projects.
- (c) liaison and coordination with industrial, academic, professional and other organizations to collaborate on projects, where appropriate.

2.2 The Joint Projects will be predominantly be undertaken in the Laboratories Canada funded facilities at both NRCan-CanmetMATERIALS Hamilton and at NRC-Mississauga Advanced Manufacturing Research.

3. GOVERNANCE

3.1 The Participants agree that each Participants efforts under this Arrangement including the selection of Joint Projects shall be subject to each Participant’s internal policies and procedures and, once established by the Participants, the governance of MADE.

4. FUNDING

4.1 The Participants are generally responsible for funding their own participation in Joint Projects.

4.2 The contributions of each Participant to a Joint Project shall be outlined in the applicable JPA.

4.3 Nothing herein obligates a Participant to participate in any Joint Project.

5. JOINT PROJECTS

5.1 The Participants shall develop and submit proposals for Joint Projects in accordance with their respective internal policies and procedures and, once established by the Participants, to the appropriate MADE sub-Committee for approval, using a proposal template format that aligns with the internal policies and procedures of the Participants and in a form that aligns with the SOWD. The Joint Project proposal shall include, at minimum, background, sections to outline objectives, project tasks, responsibilities, schedule, Deliverables, required Background IP and project budget with associated values of any in-kind and, if applicable, cash contributions.

5.2 The Participants agree to execute a Joint Project Arrangement for each approved Joint Project. Each JPA shall be signed by authorized representatives of each Participant. The JPA will become part of this Arrangement once executed by both Participants.

5.3 Each Participant shall:



- (a) Perform the work attributed to it in each Joint Project as described in the JPA SOWD. The Participants may subcontract portions of its work under the Joint Project to a third party, where such subcontracted portions will be performed in accordance with the relevant sections of this Arrangement. The Participant that is subcontracting will remain responsible for the performance of that portion of the Joint Project by the subcontractor;
 - (b) appoint a principal investigator for each Joint Project and such person shall be identified in the JPA . The principal investigators shall be responsible for overseeing the work of their respective Participants in a Joint Project; and
 - (c) make reasonable efforts to meet its delivery dates set out in each JPA SOWD.
- 5.4** The Participants agree the Joint Projects will be performed in a professional manner conforming to generally accepted scientific, technical, engineering, and environmental practices. Because of the research nature of the Joint Projects, no specific result is promised.
- 5.5** Any Tangible Property purchased by a Participant under a Joint Project shall be the property of that Participant and remain under the control of that Participant unless otherwise outlined in the applicable JPA.
- 5.6** Any Tangible Property supplied by a Participant to the other Participant for use in the performance of a Joint Project will remain the property of the supplying Participant and shall be returned to such Participant at the end of the Joint Project unless otherwise outlined in the applicable JPA.

6. INVOLVEMENT OF THIRD PARTIES

- 6.1** If both Participants wish to undertake a project under this Arrangement that will include a third party, ("**Third-Party Collaboration**"), the Third-Party Collaboration must be approved in accordance with the project selection process of this Arrangement. On approval of the Third-Party Collaboration, the Participants shall enter into a separate agreement with the relevant third party(ies). As appropriate, the Participants agree to use the one of the Participants standard collaborative research templates as the basis for the Third-Party Collaboration agreement. Nothing herein obligates such Third-Party Collaboration to be subject to the terms of this Arrangement.
- 6.2** Nothing restricts a Participant from performing research with a third party without the other Participant.

7. SECURITY OF INFORMATION

- 7.1** To assist each other in complying with all applicable statutes, regulations, and Government of Canada policies and rules concerning security and management of Canada's information, records, and resources, the Participants agree that:
- (a) Each Participant will advise the other Participant of the appropriate designation or security classification for all protected or classified information it discloses to the other Participant under this Arrangement or a Joint Project, as follows:
 - (i) protected or classified information that is disclosed in writing will be marked "PROTECTED A", "PROTECTED B", "CONFIDENTIAL", or "SECRET" as appropriate, at the time of disclosure, and
 - (ii) protected or classified information that is disclosed orally, audibly, or visually will be followed, within 20 days of its disclosure, by a written version of the information marked "PROTECTED A", "PROTECTED B", "CONFIDENTIAL" or "SECRET", as appropriate;



- (iii) any third party business confidential information that is disclosed in any format will be treated as either "PROTECTED A", or "PROTECTED B" as appropriate and shall use such business confidential information for the sole purpose of performing the applicable Joint Project;
- (b) Each Participant will advise the other Participant, in writing, of the appropriate level of protection or security classification, if any, for the information contained in its Deliverables;
- (c) A Participant, when advised of the protected or classified nature of the information that is disclosed to it or that is contained in the Deliverables of the other Participant, will safeguard that information in accordance with Government requirements for the protection or security classification identified by the other Participant. In addition to safely storing and handling such information, each Participant will limit access to such information to its, employees, agents and contractors who have the proper security clearance to access that information and a need to know.
- (d) The Participants will not exchange Protected C or Top-Secret information under this Arrangement or under any JPA.

8. INTELLECTUAL PROPERTY

- 8.1** Each Participant retains the administration and control of all intellectual property rights related to its Background IP and Solely Created Arising IP.
- 8.2** Each Participant may reproduce and use any written material contained in the Deliverables of the other Participant in fulfilment of its mandate, subject to Government requirements associated with the protected or classified markings on such material.
- 8.3** Each Participant agrees to extend the following rights of use where legally possible:
 - (a) Each Participant may use the Solely Created Arising IP of the other Participant for performing the Project and Government Purposes only.
 - (b) Each Participant may use Background IP of the other Participant for performing the Project and for Government Purposes only, and only to the extent necessary to make use of the Solely Created Arising IP of the other Participant or Jointly Created Arising IP.
 - (c) If such rights of use cannot be extended, the Participants agree to identify such limitations in the applicable JPA.
- 8.4** In the event that Jointly Created Arising IP is created during the course of the Project, the Participants agree that each Participant may use the Jointly Created Arising IP for performing the Project and for Government Purposes. Within ninety (90) days of the end of the Project, the Participants shall negotiate in good faith how to handle the Jointly Created Arising IP, including the IP administration and control of the Jointly Created Arising IP, as well as protection and licensing to third parties.
- 8.5** The Participants agree that if the nature of the Joint Project and the Background IP being utilized requires deviations from the Intellectual Property terms outlined in this article, such deviations will be mutually agreed to by the Participants as part of the Joint Project approval process and such deviations will be formalized in the applicable JPA.



9. PUBLICATIONS AND PUBLICITY

- 9.1** It is recognized that scientific publications may arise from a Joint Project. The Participants will give due consideration to the diffusion of scientific knowledge, and for that purpose may jointly publish, or jointly agree in writing to allow one Participant to publish the results of a Joint Project. Each Participant shall notify the other Participant prior to publishing any reports, articles, papers or other publications including (but not limited to) posting to a website, arising from or based on a Joint Project, and shall include a copy of the intended publication with the notification. The notified Participant will respond to the other Participant within thirty (30) days after reception of the intended publication document, if it has any concerns with publication. Each Participant will provide acknowledgement to the other Participant whenever information from the Deliverables of a Joint Project is cited in either a published or unpublished document, including internal government documents or other communications.
- 9.2** Each Participant shall acknowledge the contributions of the other Participant, including individual researchers involved, in internal and external communications as well as in publications arising from any Joint Project undertaken under this Arrangement. No Participant will publicly suggest that the other Participant endorses or recommends any product or process or results of a Joint Project.

10. ACCESS TO PREMISES

- 10.1** The Participants agree the terms outlined in this Article do not apply to NRCan's access to the NRC Building and Licensed Premises as contemplated in the License to Occupy and such access will be managed in accordance with that arrangement. For clarity, this article shall apply to NRCan's access to NRC facilities not addressed by the License to Occupy that is required for a Joint Project.
- 10.2** Each Participant shall be responsible for the actions of its employees while working in the facilities of the other Participant, and shall ensure that, while working in the other Participant's facilities, its employees comply with any and all procedures, regulations, policies and directives that the hosting Participant may adopt from time to time to address access to its facilities and activities thereon, and any and all security conditions, procedures, policies and directives that the Government of Canada may promulgate from time to time.
- 10.3** Each Participant shall provide to the other Participant access to all relevant policies, procedures and directives as well as notice of any changes, and shall provide security, health and safety training to the Participant's employees as soon as possible following permitted access to its facilities.

11. TERM, TERMINATION AND AMENDMENTS

- 11.1** This Arrangement will take effect on the date it is signed by the last Participant and will remain in effect for five (5) years or until all associated Joint Projects are completed, whichever is later, unless terminated earlier in accordance with this Arrangement.
- 11.2** This Arrangement may be terminated by either Participant upon ninety (90) days written notice to the other Participant. Upon termination of the Arrangement, all JPAs shall terminate concurrently. The Parties agree to meet and negotiate the appropriate timing and costs to wind-down any ongoing Joint Projects.
- 11.3** A JPA may be terminated by either Participant upon sixty (60) days written notice to the other Participant, in which event the Participants will take all reasonable steps to wind down the Joint Project with a minimum of costs. Termination of a JPA does not affect the ongoing validity of the Arrangement or any other JPA.



- 11.4 This Arrangement and any JPA may only be amended by written agreement, signed by authorized representatives of both Participants.

12. SETTLEMENT OF DISPUTES

- 12.1 Any disputes concerning the interpretation or implementation of this Arrangement or any JPA will be resolved by consultation among the Participants.

13. PRIMARY CONTACTS

- 13.1 Any notice related to this Arrangement will be sent to the primary contact at the addresses herein by electronic mail, which are deemed to be effective notice on the next business day.

- (a) If to NRCan, at:

Hitesh Jain - hitesh.jain@NRCan-RNCan.gc.ca

with a copy to:

Connor Leroux - connor.leroux@NRCan-RNCan.gc.ca

- (b) If to NRC at:

Dr. Caroline Cloutier - Caroline.Cloutier@nrc-cnrc.gc.ca

with a copy to:

Rob James – Rob.James@nrc-cnrc.gc.ca

A Participant may at any time change its notice contacts upon written notice to the other Participant.

14. APPROVALS

FOR THE NRC

Jean-François Houle
Vice President, Engineering

May 13, 2024

Date

FOR NRCAN

Drew Leyburne
Assistant Deputy Minister, EETS/ADMO
Natural Resources Canada

May 3, 2024

Date



ANNEX A: JOINT PROJECT ARRANGEMENT TEMPLATE

Protected [A/B](#)

JOINT PROJECT ARRANGEMENT FOR COLLABORATIVE RESEARCH IN SUPPORT OF MADE

BETWEEN: NATIONAL RESEARCH COUNCIL OF CANADA

whose head office address is:

1200 Montreal Road
Ottawa, Ontario K1A 0R6

(called the “**NRC**”)

AND: DEPARTMENT OF NATURAL RESOURCES

whose head office address is:

580 Booth Street,
Ottawa, Ontario K1A 0E4

(called “**NRCan**”)

(referred to individually as “**Participant**” and collectively as “**Participants**”)

WHEREAS NRCan and the NRC wish to enter into this Joint Project Arrangement for Collaborative Research (this “**JPA**”).

NOW THEREFORE, the Participants agree as follows:

1. GENERAL ARRANGEMENT

- 1.1** This JPA calls upon and incorporates by reference all general understanding of the Interdepartmental Letter of Arrangement for Collaborative Research Projects in the support of the Materials Acceleration Demonstration Ecosystem (MADE) (called the “**Arrangement**”), signed by the Participants on [xxx date](#).

2. PROJECT

- 2.1** This JPA concerns collaborative research, called the “**Joint Project**”, entitled [insert title](#) and further described in the Statement of Work and Deliverables attached as Schedule 1.
- 2.2** The Participants will perform the Joint Project as described in Schedule 1.

3. FINANCIALS

- 3.1** The total value of the Joint Project is estimated to be: \$[insert numeric value](#).
- 3.2** The NRC shall make an in-kind contribution to the Joint Project by performing its portion of the Joint Project at an estimated total value of \$[insert numeric value](#), as follows:
- \$[insert numeric value](#) in Government fiscal year [insert fiscal year](#); and
- \$[insert numeric value](#) in Government fiscal year [insert fiscal year](#); and
- \$[insert numeric value](#) in Government fiscal year [insert fiscal year](#).



- 3.3 NRCan shall make an in-kind contribution to the Joint Project by performing its portion of the Joint Project at an estimated total value of \$*insert numeric value*, as follows:

\$*insert numeric value* in Government fiscal year *insert fiscal year*; and
\$*insert numeric value* in Government fiscal year *insert fiscal year*; and
\$*insert numeric value* in Government fiscal year *insert fiscal year*.

Payments are unlikely in a collaborative arrangement. If no cash contributions are planned from one Participant to the other, delete sections 3.4 to 3.8.

- 3.4 *Insert acronym of paying Participant* shall pay \$*insert amount to be paid* to *Insert acronym of receiving Participant*, as follows: *Edit as needed*

\$*insert numeric value* in Government fiscal year *insert fiscal year*; and
\$*insert numeric value* in Government fiscal year *insert fiscal year*; and
\$*insert numeric value* in Government fiscal year *insert fiscal year*.

- 3.5 *Insert acronym of receiving Participant* will execute interdepartmental settlements as outlined in the following **Schedule of Interdepartmental Settlements** to advance from *Insert acronym of paying Participant* to *Insert acronym of receiving Participant* the amounts identified in the preceding section.

Schedule of Interdepartmental Settlements		
Deliverable or Milestone	Estimated Settlement Date	Settlement Amount
TOTAL		

- 3.6 The Participants are Financial Interdepartmental Settlement (FIS) compliant. The FIS coding information for *Insert acronym of paying Participant* is: *Insert details if known at time of drafting*

Intra No. _____
IS Org. No. _____
IS Ref. Code _____

- 3.7 FIS Transaction documents will be sent by *Insert acronym of receiving Participant* to *Insert acronym of paying Participant* to the attention of its FIS representative identified in the next section.

- 3.8 The FIS representative for NRCan for the purpose of this JPA is: *Insert details if known at time of drafting*

Name: _____
Tel No.: _____
e-mail: _____

The FIS representative for the NRC for the purpose of this JPA is:

Name: *insert name of NRC financial officer for the NRC host program*



Tel No.: *insert telephone number of NRC financial officer for the NRC host program*

e-mail: *insert email address of NRC financial officer for the NRC host program*

- 3.9** Each Participant will notify the other Participant and explain, if it appears at any time that the values it expressed as an estimate will be exceeded. The Participants will suspend performance of the Joint Project pending a mutual decision, in writing, by the Participants to modify or discontinue or continue the Joint Project.

4. TERM AND TERMINATION

- 4.1** This JPA takes effect on the date when the last Participant has signed and shall remain in effect until *insert date*, conditional upon all Deliverables having been delivered by the Participants, unless terminated earlier.

5. DEVIATIONS

- 5.1** Tangible Property: *If any deviations from sections 5.5 and 5.6 related to, Tangible Property, in the Arrangement have been agreed to, such should be noted here. Otherwise, note "Not applicable".*
- 5.2** Intellectual Property: *If any deviations from article 8, Intellectual Property, in the Arrangement have been agreed to, such should be noted here. Any limitations to note as per section 8.3(c) or 8.5. Otherwise, note "Not applicable".*

SIGNED by NRCan

DEPARTMENT OF NATURAL RESOURCES

Date: _____
(yyyy-mm-dd)

Per: _____

Name: _____

Title: _____

SIGNED by the NRC

NATIONAL RESEARCH COUNCIL OF CANADA

Date: _____
(yyyy-mm-dd)

Per: _____

Name: _____

Title: _____



SCHEDULE 1: STATEMENT OF WORK AND DELIVERABLES

Objectives

Identify the goals of the Joint Project.

Work Plan

List and describe the tasks to be performed. For simple tasks, listing steps in bullet form is acceptable and preferred. If any task is optional, or dependant either on the deliverables of the previous task(s) or on a Participant's decision, please clearly flag and describe this in this Work Plan section.

For each task, indicate which Participant is responsible for its performance. If any task appears to be the responsibility of more than one Participant, it must be further divided into sub-tasks until only one Participant is responsible for each specific sub-task and associated deliverable.

Prerequisites for Task Completion

If either Participant must provide anything to the other (for example, any information or any samples, equipment, materials or tools, any necessary approvals, etc.) for the first Participant to adequately perform any task, please indicate and describe these obligations here. Clearly define the dates when the equipment, prototypes, samples, or any other materials or tools must be made available to the participant performing the work. Make it clear which Participant is responsible for all costs for logistical arrangements to and from the other Participants facilities (e.g., shipping). Include post-project considerations as applicable, such as return or disposal of material.

Task Expected Deliverables

Clearly state all targeted or expected deliverables and which Participant is responsible for that deliverable. Include acceptance criteria if applicable.

Estimated Timeline

Indicate an estimated start date and duration of the Joint Project. Gantt Charts or other project timeline methods can be used here.

If it is not possible or not practical to indicate timelines for each task, please indicate only the estimated overall project duration.

Assumptions, Constraints and Risks

Please delete header & this entire paragraph if it is not applicable.

Document here any considerations on which the Work Plan is based, such as assumptions and constraints, as well as any known risks associated with the Joint Project.

Background Intellectual Property

- Identify any background IP (BIP) (including any know-how) of each Participant that will be required to carry out the proposed work of both the NRC and NRCan. Indicate "none" if none is being used.*



BIP of the NRC

Insert. Indicate "none" if none is being used.

BIP of NRCan

Insert. Indicate "none" if none is being used.

JOINT PROJECT CONTACTS

Role	Name	Phone	e-mail
NRCan Primary Investigator	<i>INSERT TITLE AND NAME</i>	<i>(XXX) XXX-XXXX</i>	<i>Insert e-mail address</i>
NRC Primary Investigator	<i>INSERT TITLE AND NAME</i>	<i>(XXX) XXX-XXXX</i>	<i>Insert e-mail address</i>



ANNEX B: SCIENCE STATEMENT OF REQUIREMENTS (SSoR)



LABORATORIES

CANADA

Terra Canada Mississauga / Hamilton Centres of Expertise (Mississauga / Hamilton)

Science Statement of Requirements

November 5, 2019







Table of Contents

Executive Summary	4
1 Introduction	6
1.1 Problems and Opportunities.....	6
1.2 Previous Approvals.....	6
2 Background/Strategic Context	6
2.1 Laboratories Canada Program	6
2.2 Portfolio Science Strategy.....	7
3 Project Scope/Client Requirements	8
3.1 Science Focus by Project.....	8
3.2 Space Typology Requirements.....	11
3.3 Personnel Considerations	15
3.4 Consolidation Strategy	15
3.5 Equipment Strategy	18
3.6 IM/IT Strategy	20
3.7 Security	22
3.8 Site Considerations	23
4 Risks	23
5 Source of Funds	25
6 Approvals	25
6.1 Approval of this Science Statement of Requirements and to Initiate Further Documentation .	25
APPENDIX A Space Typology Definitions	26
APPENDIX B Science Plan	28
APPENDIX C Hamilton Business Case	32



Executive Summary

Section 1 - Introduction: The proposed Terra Canada Mississauga / Hamilton Centres of Expertise (Mississauga / Hamilton) will form, in Canada's southwest Ontario industrial heartland, a world-class collaborative research capability built around new innovative artificial intelligence (AI) driven, computer-enabled robotics technology to accelerate discovery and innovation of new materials technologies. These Centres will directly align to Terra Canada's Low Carbon Economy science theme, and support both the Sustainable Lands and Resource Development and the Health and Safety science themes, and provides an opportunity to establish the first-of-its-kind critical mass of expertise in Canada to accelerate discovery and development of advanced materials, particularly in the clean energy domain. This new automated discovery capability will directly address the notoriously slow and expensive Edisonian process of discovery and innovation of the underpinning new materials that are urgently needed to support the nation's future low carbon economy. Ultimately, such a strategy will result in fast-tracked materials innovations that will redefine the manufacturing and energy industries with novel processes and products, while meeting the sustainability, productivity, quality and cost reduction demands of competitive global markets.

Section 2 - Background / Strategic Context: The initiative will develop a new laboratory within Natural Resources Canada's (NRCan) CanmetMATERIALS (CMAT) facility in Hamilton (approved by Terra Canada governance) and expand the National Research Council's (NRC) new, under-construction, facility in Mississauga (NRC-M). This will integrate the substantial materials science and engineering capabilities of both organizations with those of a broad range of academic, OGD and industry partners across Canada in order to create a unique new set of science platforms and a national research network serving the clean energy, transportation, manufacturing and other industry sectors.

Section 3 - Project Scope/Client Requirements: This initiative will develop and deploy a set of integrated science and technology platforms via 3 core Science Programs: 1) *Accelerated Materials Discovery and Process Development*, 2) *Materials Production Scale-up, Demonstration and Standardization*, and 3) *Materials and Process Sustainability and Safety*. Key materials of focus will include materials for clean energy, novel materials for additive manufacturing, and advanced "multi-functional" materials. In addition to NRC and NRCan, this will also integrate the efforts of a broad range of collaborators across the national innovation system including the Universities of Toronto, McMaster, Waterloo and British Columbia, Xerox Research Centre of Canada, the NGen Supercluster and the Canadian Institute for Advanced Research as well as Lawrence Berkeley National Laboratory and other international partners.

Establishment of these capabilities will require renovation of approximately 91.5 m² laboratory within CMAT (Hamilton) to allow work to begin as soon as possible to develop Materials Acceleration Platforms (MAPs), the AI / robotics technology that will form the centrepiece of Science Program 1. When complete, the proposed Terra Canada expansion of NRC-M (Mississauga) would play host to a much larger set of MAPs, largely open to use by Mississauga / Hamilton's national network of partners, as well as the complementary activities of Science Programs 2 and 3.

It was estimated during the Memorandum to Cabinet process that 40 FTEs from NRC would be located in the expanded NRC-M location. Following strategic discussions with NRCan and partners, this number is estimated to grow to 66 FTEs in the expanded NRC-M facility in order to accommodate NRCan and visiting collaborators. CMAT will also host NRC and visiting researchers, but these will be accommodated within existing facilities.

A broad range of existing and new equipment and facilities will be integrated within Mississauga / Hamilton in support of each of the 3 Science Programs. In keeping with its highly collaborative nature, equipment owned by both NRC and NRCan will be located in the facilities of the other partner as will



critical equipment owned by other partners and collaborators. The nature of the work to be undertaken as part of this initiative means that a world-class, high-performance, flexible, upgradeable and secure IT / IM system will be absolutely fundamental to its success.

Section 4 - Risks: Most of the risks associated with establishment and operation of the proposed initiative are similar to those envisioned for other Laboratories Canada initiatives, such as those associated with securing building capital and effectively managing site(s) construction. Unusually high risks in this case relate to the criticality of high-performance IT / IM (assessed as Very High risk) and securing remaining needed capital funding for acquisition of equipment (assessed as High risk).



1 Introduction

1.1 Problems and Opportunities

The Greater Toronto and Hamilton Area is home to a robust and diverse advanced manufacturing sector that includes aerospace, automotive, food and beverage and clean technologies. Recent investments by Ontario (\$35M for the Advanced Manufacturing Consortium linking Hamilton, Waterloo and London) and the Federal government (\$200M for the Advanced Manufacturing Supercluster) will build on the innovation strengths of southern Ontario to help manufacturers across Canada lead in application of advanced technologies.

Adding to the momentum, NRCan CanmetMATERIALS (CMAT) laboratory in Hamilton, and NRC's new Advanced Materials facility being built in Mississauga (Phase 1 of NRC-M) are ideally located to create and catalyze rapid growth through discovery, development and application of new materials.

Now is the prime opportunity to leverage and integrate the complementary expertise of CMAT and NRC-M and expand the specialized experimental spaces into modern facilities that will contribute to federal scientific excellence, enhance collaboration, and accelerate innovation for advanced materials development particularly in the clean energy domain.

Terra Canada is the R&D Cluster that addresses the Laboratories Canada vision of providing transdisciplinary R&D and Innovation for Sustainable Land and Resource Development, Low Carbon Economy and Health and Safety for Canadians. Directly aligned with Terra Canada's Low Carbon Economy science theme and supporting both the Sustainable Lands and Resource Development and the Health and Safety science themes, NRC and NRCan aim to quickly realize a highly collaborative Mississauga / Hamilton Centres of Expertise (Mississauga / Hamilton) in accelerated materials discovery and innovation in the manufacturing heart of southwest Ontario, building on and leveraging existing and in-construction facilities of both organizations in the region.

1.2 Previous Approvals

The February 2018 Budget announced significant new investments to support the renewal of federal science by launching the first phase of an ambitious plan to rebuild federal laboratories. Budget 2017 highlighted the importance of federal science by its commitment to develop a federal science infrastructure strategy based on a more integrated, effective approach to build world-class, multi-departmental collaborative facilities. In November, 2018, the Government approved \$2.8B in funding Laboratories Canada, previously called Federal Science and Technology Infrastructure initiative (FSTII). Laboratories Canada is an ambitious whole of government 25 year transformative strategy, signalling a new and forward looking approach to science planning and collaboration. The first installation (2018 – 2023) of Laboratories Canada addresses the federal science facilities in the most critical condition and exploitation of near-term opportunities for leveraging existing initiatives. The May 2019 Treasury Board (TB) Submission, approved, will provide funding towards Terra Canada cluster planning and early design. Laboratories Canada will act as project sponsor and be responsible for delivery of the new facilities in collaboration with the cluster.

2 Background/Strategic Context

2.1 Laboratories Canada Program

Laboratories Canada building projects will be carried out with direct support and funding from Public Service and Procurement Canada (PSPC).

Six principles guide the initiative: science excellence, collaboration, a diverse and inclusive talent pool, agility and responsiveness, environmental responsibility, and responsible public stewardship. In addition



to fostering collaborative science, Laboratories Canada is leading a shift to a whole of government approach to the science and technology fixed asset class. The approach includes four interdependent components that will enhance federal S&T capacity, address infrastructure deficiencies and enable collaboration and cutting-edge science by:

- Building / integrating new multi-department, multi-purpose, “green” federal laboratories;
- Upgrading S&T IM/IT systems to facilitate data-sharing and high-capacity computing while ensuring security of government systems;
- Optimizing the impact of investments through shared major scientific equipment; and
- Reducing policy barriers that inhibit scientific collaboration.

2.2 Portfolio Science Strategy

NRCan’s CMAT science facility, commissioned in 2010 as LEED Platinum, is located in Hamilton’s McMaster Innovation Park adjacent to McMaster University. Situated ~40 km away in the Sheridan Science and Technology Park is the NRC’s new Mississauga research facility (NRC-M), with Phase 1 presently under construction to establish the first floor for Advanced Materials Manufacturing.

This proposals is to integrate the expertise of NRC and NRCan, along with a range of regional and national academic and industry partners in these regions by:

- 1) Expanding NRC-M construction project beyond the first floor.
- 2) Renovating existing CMAT space to add new capabilities,

NRC and NRCan have collaboratively developed strategic initiatives in advanced clean energy materials and integrated plans to substantially accelerate materials discovery and development by deploying a set of cutting-edge technologies including artificial intelligence (AI), robotics and high performance computing. This automated approach has been said to represent a “second revolution in science” and will be realized through the development of integrated, open-access platforms known as Materials Acceleration Platforms (MAPs). NRC and NRCan are now poised to leverage the decision-making strengths of AI and new efficiencies to experimentation never before possible to bring the breadth and depth of the advanced materials science portfolio to new levels. Mississauga / Hamilton will couple this novel approach to materials discovery with complimentary technologies and platforms that address critical materials development and commercialization challenges, such as production scale-up, health, environmental and life-cycle concerns to accelerate the transition to industry and reduce time to market.

3 Project Scope/Client Requirements

3.1 Science Focus by Project

The Terra Canada Low Carbon Economy theme will enable a first-of-its-kind Canadian network, with researchers in the **NCA working closely with** Terra Canada regional locations in **Mississauga and Hamilton** (competencies in Accelerated Materials Discovery, Production Scale-up; Standardization; and Materials and Process Sustainability and Safety) and **Winnipeg** (competencies in Sustainable Materials Manufacturing, and Additive and Digital Manufacturing). The network will expand beyond Terra Canada labs – seeking to establish knowledge sharing with air quality/climate modelling R&D in Montreal/Victoria, and advanced materials and manufacturing R&D in Boucherville and Montreal, and energy R&D in Vancouver, Devon, and Varennes, as well as ECCC’s air quality/climate modelling in Montreal/Victoria.

The Terra Canada Mississauga / Hamilton region will develop and deploy a set of integrated technology platforms to accelerate discovery of new materials and processes which will underpin commercialization of disruptive new products in support of Canada’s future low-carbon economy (Figure 1-1)



Figure 3-1. Mississauga / Hamilton Centres of Expertise

This new collaborative capability will focus on resolving a fundamental challenge in bringing to market urgently-needed new technologies to address climate change and other environmental problems: the slow pace of discovery and development of the new materials at the very heart of these new technologies. This problem will be addressed through application and integration of potentially revolutionary new developments in artificial intelligence and robotics-based materials discovery with existing state-of-the-art materials science facilities and programs at NRC, NRCan and a range of world-class regional, national and international partners. The planned joint science program will also leverage a host of on-going initiatives such as NRC’s new Challenge Program in Materials for Clean Fuels, the NRC-UofT Collaboration Centre in Green Energy Materials, NRCan’s investment in University of British Columbia’s (UBC) project Ada and CMAT’s science program on clean energy materials. It will also be closely coordinated with other



regional and national Government of Canada investments in the NGen (Next Generation Manufacturing) and Scale.AI superclusters, and the PanCanadian AI Strategy.

As the hub of a regional and national innovation network in accelerated materials discovery, the proposed new initiative has tremendous potential to integrate Canada's exceptional competencies in materials science to rapidly deliver solutions to the nation's urgent environmental challenges.

The new joint endeavour will bring together Canada's top researchers in materials science and engineering from across the nation both to create and implement its unique suite of AI, robotics and other technology platforms and to exploit these new platforms to discover, develop and deploy disruptive new materials. These platforms will be developed and applied as part of the three core programs:

Science Program 1: Accelerated Materials Discovery and Process Development

Issue:

The innovation process for development of new materials, from laboratory to industrial application, is currently expensive and slow, often requiring 20 years for full realization. With Terra Canada Mississauga/Hamilton, there is a considerable opportunity to build on Canada's strong research capabilities in materials science via the application of new approaches in AI-driven, computer-enabled (smart) robotic platforms. These platforms will combine high-throughput experimental methods and new AI techniques to accelerate discovery and development of novel materials for clean energy production and use. They will impact energy conversion, storage and distribution, making Canada a leader in commercialization of these and other advanced materials.

Methodology:

Through close collaboration with the international Mission Innovation Clean Energy Materials - Innovation Challenge 6 (<http://mission-innovation.net/our-work/innovation-challenges>), which Canada (via NRCan) co-leads, and world-class regional, national and international partners, Mississauga / Hamilton will facilitate development and application of a suite of technologies associated with AI-driven, smart robotic platforms – called Materials Acceleration Platforms (MAPs). Based on a common framework of computational, experimental and automation technologies, a series of MAPs will be deployed to accelerate discovery of organic, inorganic, functional and structural materials for a range of applications. MAP design and operation will follow the approach under development in the NRCan-funded UBC project Ada (Autonomous Discovery Accelerator, <http://www.projectada.ca/>), a first-of-its-kind demonstrator.

Outcomes:

- Establishment of an accessible national platform enabling acceleration of the rate of discovery of new materials for clean energy applications and other areas by a factor of 10 (e.g. from 20 years to 2 years)
- Development of a pipeline of new high-value materials available for transition to regional and national industrial innovation partners, enabling products supporting substantial reductions in CO₂ emissions

Science Program 2: Materials Production Scale-up, Demonstration and Standardization

Issue:

Discovery of new and potentially disruptive materials, and development of processes for their synthesis are but the first steps on the pathway to commercialization and application. Accelerated materials discovery is of limited value if these materials do not ultimately meet the performance standards needed for real-world applications and cannot be consistently produced at high quality, low cost and in sufficient quantities to meet market demands.

Methodology:

In parallel with development of AI-driven, smart robotic experimental platforms that will comprise the visionary new core capability of the Mississauga / Hamilton, Science Program 2 will develop a set of



complementary technologies in materials and process standards development, scaled-up materials synthesis and production, and design, prototyping and demonstration of small-scale devices and systems. These technologies will be developed and deployed together with Canadian and international partners, including leading SMEs, multinationals, and standards-development organizations (e.g. ASTM, NRC Metrology, NIST), and will support both “conventional” routes for materials technology development / transition and establish a suite of automated tools that will be integrated into MAPs.

Outcomes:

- Establishment of a set of accessible, collaborative platforms and standards supporting production scale-up, demonstration of new materials for applications in a wide variety of areas including clean energy, and other areas enabled by additive and smart materials
- Maturation, productionization, demonstration and transition to industry of new innovative materials, processes, devices and systems, creating economic value for Canada

Science Program 3: Materials and Process Sustainability and Safety

Issue:

Assessment of the long-term durability and performance of novel materials in real-world use cases, and the impact of the production, use and end-of-life disposal of these materials on human health and the environment, including greenhouse gas (GHG) emissions, are often neglected in materials research, discovery and development. As a result, many initially-promising materials have proven to be dead-ends or require lengthy and expensive re-engineering to mitigate issues such as durability, environmental or health impacts that could have been anticipated and avoided had they been considered early in the development process (e.g. BPA, asbestos, single-use plastics, etc.).

Methodology:

Science Program 3 will develop and apply a set of technologies and standards relating to materials and devices ageing and durability and the impact of their production, use and disposal on health and the environment. Specifically, these platform technologies will include a set of protocols, experimental and analytical tools to address:

- Accelerated ageing of materials and devices, combining physical testing with predictive modeling
- Cradle-to-cradle materials and product life cycle assessment (LCA), coupled with technoeconomic analyses (TEA) and materials resource efficiency modelling
- Health and environmental impacts and methods for their mitigation

As with the other two Science Programs, development and deployment of these platforms will be undertaken in collaboration with a range of partners and will both support conventional materials development routes and be developed for direct integration into MAPs. These platforms will be designed for open access to collaborators and other Canadian researchers and materials innovators.

Outcomes:

- Establishment of a set of accessible, collaboration oriented platforms supporting assessment of materials and device durability and impact on health and the environment
- Transition to industry partners of a range of durable new materials and products which, by displacing other materials and products, create a net positive impact on Canadian health and the environment
- Development of standards for materials quality, production, and life-cycle management

Science Program Application Areas

While the materials and process development platforms established as part of the three Science Programs can in principle be applied to nearly any type material or application, practical and resource considerations



will drive a focus on materials and applications with potential for substantial impact on Canada's future low-carbon economy. These will include:

- Materials for clean energy, including energy conversion, storage and distribution. Initial focus areas will be functional materials: thermoelectrics for conversion of waste heat to electricity, high-entropy alloys for extreme environments, catalysts for conversion of CO₂ to fuels and other hydrocarbons, and low-cost, renewable hydrogen generation
- Materials for additive manufacturing processes, supporting improved materials utilization, enhancing recycling, higher performance and improved design / more efficient vehicles and energy transformation systems
- Advanced multi-functional materials¹ for printed electronics and wearables, aiming at durable, energy efficient and recyclable / biodegradable materials and products

Science Program Partners

The breadth of competencies, the range of equipment, and the quantity of resources required to establish and implement the complex platforms described in this section cannot be supported by Mississauga / Hamilton alone, even by exploiting the considerable combined materials science and engineering competencies of NRC and NRCan. The described vision will only be possible by establishing Mississauga / Hamilton as an initial "critical mass" and focal point of a national network, integrating the efforts of collaborators across Canada, with contributions from select international partners.

As described in Section 3.3, the growing list of partners for Mississauga / Hamilton include many of Canada's most prominent players in materials science and innovation including the University of Toronto (UofT), UBC, the University of Waterloo (UWaterloo), McMaster University and Xerox Research Centre of Canada (XRCC). UBC and UofT already have world-class and growing competencies in AI-based accelerated materials discovery, with UWaterloo, McMaster University and XRCC having world-class competencies in, respectively, AI, materials characterization and translation of materials discoveries into innovative products, and all can benefit substantially from pooling of competencies and resources toward this new challenge.

3.2 Space Typology Requirements

The Terra Canadab proposal is for two integrated locations with staff and equipment from each organization co-located. This section describes space requirements for re-purposing existing space at CMAT (Hamilton – see Appendix C) and expanding NRC-M (under construction in Mississauga)². The project currently underway at NRC-M³ is Phase 1 (NRC funded), while Phase 2 represents the proposed Laboratories Canada expansion. Hence:

- The site and constraints of the 2 building locations are known. The yellow arrows in Figure 3-2 and Figure 3-2**Error! Reference source not found.** show the 0.94 hectare site acquired by NRC from Xerox in November 2017 and the gross area of NRC-M, respectively. The maximum site building capacity is 5614 m². Figure 3-4 highlights the area to be re-purposed in CMAT.
- There is supporting infrastructure within the buildings that will serve the new (Phase 2) space such as, lobbies, loading/storage areas, characterisation labs, and parking.
- Estimates have been provided by Diamond Schmitt Architects (DSA) - the original designer of both

¹ Materials that perform multiple functions in a system. For example, a material that combines high mechanical strength while also being able to absorb energy.

² Establishment of proposed TerraCanada facilities in Hamilton and Mississauga can launch more quickly than a "green field" proposal since, for example, sites are established, permits are not required, and design considerations have been accounted for.

³ Represents completed 1st and shelled 2nd floors. Completion date: December 2019 - January 2020.

buildings. These will be provided to the A&E firm selected by PSPC for the Laboratories Canada program.



Figure 3-2. Site location of NRC facility in Mississauga

Table 3-1 lists space available at NRC-M from Phase 1. It also provides new lab, office, and some ancillary requirements for Mississauga / Hamilton portion of the Terra Canada science plan (Phase 2). For Phase 2:

- CMAT requires 91.5m² renovations to existing space for 2 MAP labs⁴. All ancillary space is accounted for in their existing building. A Class C estimate, was prepared in a DSA report dated April 15, 2019.
- For NRC-M, space requirements estimates are provided in table 3-1. These will be further refined during development of the feasibility report and functional program.
- The total area, expected to range between 2400-4800m² will be decided upon following cluster review of a feasibility report / options analysis, **with consideration of the current budget and contracting authorities** (for design and construction management contracts) received from Treasury Board. Framework Partners has been engaged to develop this report. The results of the options analysis and feasibility report and selection of the preferred option will go through the LC governance process for approval.
- NRC commissioned a Class D estimate for a basic functional program which was provided by DSA on February 25, 2019.
- The plan also calls for a clean room (Class 1000). The space for it is not included below as it already exists on the 1st floor. However, it will require air filtration and temperature / humidity control to be added as part of Phase 2. This 1st floor clean room, along with the materials characterisation and high bay areas established as part of the current Phase 1 construction will serve the Laboratories Canada facilities established at NRC-M.

⁴ This allows head start on research program instead of waiting until NRC-M is ready. Staff from NRC and NRCan will initially be co-located at CMAT with some equipment and staff moving to NRC-M upon Phase 2 completion.



Table 3-1. Lab Facility Space Typology Mississauga / Hamilton

Mississauga (NRC-M) / Hamilton (CMAT)				
All entries in m ²	Phase 1 (NRC-M)		Phase 2 (LC)	
Type of space	1 st floor	Fit-up (2 nd fl. shell)	Area for costing ⁵	New fit-up ⁶
A. Lab Area				
Characterisation lab side/wet dry lab	233			
Clean room lab side/wet dry lab	216			
High bay	172			
Dry laboratory			400	
Wet laboratory (CL1 with humidity control)			500	91.5
Wet laboratory (CL2):			500	
Clean room (Class 1000)				140
Sub Total	621		1400	131.5
B. Office area	378		800	
C. Common area	304		TBD	
D. Loading and storage				
Storage (flammable liquid, gas-chemical, oxygen, bicycles)	63		TBD	
Loading	62			
Light industrial/general storage (e.g. material samples)			300	
Sub Total	125		TBD	
E. Circulation area				
Stairs, elevator, corridor	346		TBD	
F. Gross Up				
Envelope, mechanical shafts	124		TBD	
G. Mechanical and electrical rooms				
Mechanical room	31		TBD	
Electrical room	31		TBD	
Sub Total	62		TBD	
C. Support spaces				
Machine shop	62			
Assembly shop			TBD	
IT Server/Computer room	24		40	
Sub Total	86		TBD	
Total floor area	2046	1850 (est)	TBD	TBD

⁵ All "TBD" entries to be determined in conjunction with the A+E firm during detailed design phase.

⁶ The clean room is located in NRC-M (1st floor). The CL1 wet lab is located in CMAT.



Figure 3-3. Footprint of NRC building under construction in Mississauga

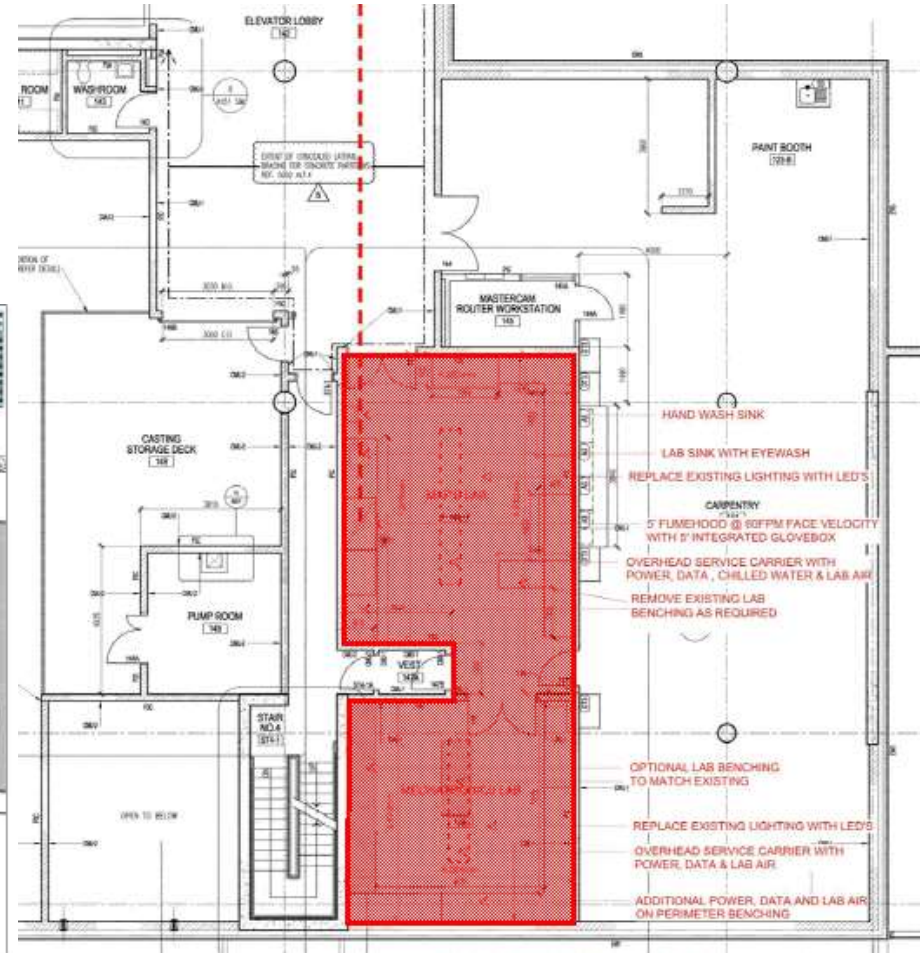


Figure 3-4. Layout of repurposed area in NRCan CMAT facility (Hamilton)



3.3 Personnel Considerations

It was estimated during the Memorandum to Cabinet process that 40 FTEs from NRC would be located in the expanded NRC-M location (Phase 2). Following strategic discussions with NRCan and partners, this number may grow to an estimated 66 FTEs in the expanded NRC-M facility to enable space for NRCan and visiting collaborators. This will be validated through the functional programming process. Any additional requirements beyond the 40 FTE will need to be approved through the cluster governance.

CMAT will also play host to visiting workers from NRC and universities, OGDs and industry, however this will be accommodated within existing space.

Table 3-2. Scientific Personnel requiring incremental office space at NRC-M by Science Program

Scientific Program		NRC Staff	NRCan Staff	Visiting Workers
#	Name			
1	Accelerated Materials Discovery and Process Development	15	10	10
2	Materials Production Scale-up, Demonstration and Standardization	15	0	3
3	Materials and Process Sustainability and Safety	10	0	3

It may be noted the above Phase 2 scientific personnel will be complemented with NRC staff and visiting workers who will occupy a total of 33 office spaces (20 NRC, 13 visiting workers) on the first floor (Phase 1) of NRC-M.

Various types of workspaces required will depend on need. These are expected to be divided into:

- Enclosed office space for staff overseeing business / personnel sensitive information
- Supervisor work space for staff needing space for staff discussions
- Scientific / technical work space for staff without special needs
- Hoteling work space for students and others needing temporary work space

3.4 Consolidation Strategy

3.4.1 Collaboration

As described earlier, this initiative will establish Mississauga / Hamilton as a first-of-its-kind capability in Canada focused on accelerating discovery and development of a wide variety of advanced materials, including the clean energy domain. This is a natural extension of current (CMAT) and planned (NRC-M Phase 1) materials discovery, design, synthesis, characterization, scale up and application capabilities; these existing competencies will be heavily leveraged as they are applied to materials that enable Canada's future low-carbon economy. Staff from NRCan and NRC will work in both facilities, developing strong synergies and cross-functional training to build unique capabilities across the three core science programs. Phase 1 of NRC-M will establish NRC's traditional synthesis and characterization tools and basic infrastructure underpinning Science Programs 2 and 3. Science Program 1 activities will launch with the completion of the renovation of CMAT's laboratory which will take place coincident with Phase 2 construction at NRC-M. A team of NRC staff will work at CMAT to support the NRCan-led development and application of MAPs. Upon completion of Phase 2 (NRC-M expansion), NRC staff and NRCan staff along with visiting workers will begin occupying this new space and together develop additional MAPs, advance the foundational capabilities established in Phase 1 to integrate with the MAPs and expand these to meet Science Program 2 and 3 outcomes. It is understood that under Terra Canada, the building will operate as



a single entity and enable collaboration. A joint NRC/NRCan presence will be maintained at both the NRC-M and CMAT as well as reciprocal access to each other's facilities. Mutual interests in materials innovation and support for advanced manufacturing have formed the basis for longstanding partnerships between respective research groups (CMAT and NRC in Saguenay (Aluminium Technology Centre), Boucherville (industrial materials) and London (advanced machining)). These collaborations in the area of materials for lightweight, fuel-efficient vehicles offer a model for maximizing impact on industry and society that can be applied to Mississauga / Hamilton.

As the locus of a national network for accelerated materials discovery and development, Mississauga / Hamilton will integrate and exploit public and private sector R&D investments, competencies and initiatives in accelerated clean energy materials discovery and development, partnering and collaborating extensively with Canada's world-class materials R&D leaders. As this initiative progresses, an engagement strategy will be developed. As this initiative grows, there may be a need for future collaborative space and/or expansion. Ongoing initiatives provide mechanisms to incite collaboration, and include: the NRCan-led Mission Innovation Clean Energy Materials Innovation Challenge (IC6), including NRCan contributions to the European Commission's Flagship Energy Technology program proposal Battery 2030+, est. 1B€; the NRC Challenge programs in Clean Energy Materials and Artificial Intelligence for Design (AI4D) which each offer >\$2M/yr (7 years) for university and SME partners; the ISED-funded NGen Advanced Manufacturing Supercluster with specific interests in additive manufacturing and for which NRC and NRCan will provide support. Joint efforts will extend or build upon existing strategic partnerships, such as:

- NRC's new Collaboration Centre in Green Energy Materials (CC-GEM) with the *University of Toronto*. Headquartered at NRC-M, CC-GEM will support NRC's Clean Energy Materials and AI4D Challenge Programs and contribute to Mission Innovation IC6. Research will focus on developing and demonstrating emerging green energy materials for next generation photovoltaics, renewable carbon-based feedstocks, renewable fuels and fuel cells, and consumer-scale energy solutions. Cross-cutting platform technologies for accelerated materials discovery and process development, production scale-up, demonstration and standardization and materials and process sustainability and safety will be developed to enable more rapid technology development and application.
- Located adjacent to the NRC's Mississauga facility and accessible through a collaboration with NRC, *Xerox Research Centre of Canada's (XRCC)* materials design, analysis, scale-up and pilot plant facilities are critical enablers to rapidly and cost-effectively scale production of material discoveries from laboratory quantities to volumes needed to advance past proof-of-concept. Moreover, NRC and XRCC are founding partners of the Canadian Campus for Advanced Materials Manufacturing (CCAMM). CCAMM is a flagship hub for sharing competencies and equipment/facilities from industry, public, and academic organizations to de-risk development and deployment of viable materials technologies and bridge the gap between laboratory demonstration and industrial scale manufacturing.
- NRC's collaboration with the *University of Waterloo* Multi-Scale Additive Manufacturing lab to develop precursor materials for additive manufacturing, in support of NRC's Advanced Manufacturing Program and the NGen Supercluster. Research will centre on understanding the influence of variations in properties of materials feedstock for additive manufacturing (AM), to lower the cost of material feedstock and develop new recycling strategies, thereby realizing uniformly high quality, novel, and lower cost AM-derived products. This collaboration will establish technical competencies in metal powder based synthesis and characterization, develop the understanding of the metal powder feedstock used in AM to create an industrially robust process, as well as lower the economic barriers to metal AM by increasing the accessibility of low-cost materials and processes.
- NRCan's long-standing collaboration with *McMaster University* in developing advanced materials, advanced imaging, and material characterization is a model for an NRC relationship with McMaster in the same areas. Imaging and material characterization inherently generates mass quantities of data



that must be processed efficiently. AI and machine learning algorithms will play an important role in unlocking the value of such data.

- NRCan's on-going partnership with *UBC* and the University of Toronto in the area of AI, advanced robotics, and materials for clean energy applications in support of Mission Innovation, including the NRCan-funded project Ada, valued at \$8M. NRC also aims to become a core partner, and has initiated collaboration with the same University of Toronto partners in CC-GEM. Project Ada offers two critical pieces to the Mississauga / Hamilton Science Program: a proven pathway and lessons learned for implementing MAPs, and a vital understanding of the technical issues underpinning future MAP installations.
- NRCan and NRC's collaborative efforts with the *CIFAR*, and collaboration with *Vector Institute* under the PanCanadian Artificial Intelligence Strategy. Materials discovery, benefits from the intellectual assets involved in CIFAR's relevant research programs including Bio-Inspired Solar Energy and Quantum Materials. Though a number of existing CIFAR fellows are involved in the Mississauga / Hamilton efforts, considerable leverage is possible through the extended networks inside and outside Canada. The collaboration with our AI institutes, including Vector, will bring greater strength and depth to accelerating materials discovery and development.
- NRCan, through *Mission Innovation* IC6, has developed an international presence and coordinates activity with Mission Innovation member countries. Most notably, the European Union Frontier Energy Technology flagship programs include substantial investment in accelerated materials discovery. Further collaborative activities are developing in Scandinavia, Germany and the UK, while Italy has a strong interest in collaboration with NRCan's first MAP installation for thermoelectrics.
- The NRCan Memorandum of Understanding (MOU) with the Materials Genome Laboratory, *Lawrence Berkeley National Laboratory* (LBNL) and developing relationships with *Lawrence Livermore National Laboratory* (LLNL) and the *National Institute for Standards and Testing* (NIST).
- The NRCan collaborative efforts with the *University of New Brunswick* and *Dalhousie University* on Additive Manufacturing under the Industrial and Technological Benefits (ITB) program under Canada's Defence Procurement Strategy.

Additional engagements with academia, OGDs, companies across the materials value chain and suppliers will continue to be explored.

3.4.2 Access to Highly Qualified Personnel (HQP)

The Mississauga / Hamilton initiative will be supported by addition of new staff and relocation of existing staff. These will be augmented by co-located visiting workers from partner organizations contributing complimentary capabilities and expertise, multiplying impacts and facilitating cross-pollination of ideas, knowledge and innovative advances/ concepts.

NRC-M and CMAT are ideally located in Canada's manufacturing heartland with 7 universities and 7 colleges nearby (<100 km), offering access to a large and diverse pool of HQP. This was apparent during a recent hiring campaign where NRC received over 300 applications for two research positions for Mississauga. The unique network, the concentration of academic resources, the new state-of-the-art facilities, leading-edge research program, and highly integrated collaborative environment will further serve to attract and retain HQP.

3.4.3 Access to Existing Knowledge

Mississauga / Hamilton will leverage and build upon on extensive existing materials science competencies across NRCan and NRC, further extending these through partnerships with leading R&D groups and industry. Access to existing knowledge will be enabled by coordinated efforts on intersecting R&D initiatives (e.g., Mission Innovation, Energy Materials and AI4Design Challenge Programs), strategic

partnerships founded on collaboration (e.g., CC-GEM, CCAMM, University of Waterloo) and co-location of NRC, NRCan and partner staff.

3.5 Equipment Strategy

The Mississauga / Hamilton facility will integrate existing equipment at CMAT with new equipment currently being procured for installation on the first floor of NRC-M by NRC and its partners at the Universities of Waterloo and Toronto and with future equipment that will be acquired for the Mississauga / Hamilton facility. The following table outlines both existing and future relevant equipment planned to be on-site at both Terra Canada locations.

Table 3-3. Equipment Strategy

Hamilton - Current Relevant Equipment					
Facility/ Functional Asset	Description		MAPs	Scale-up, Demo, Standardization	Sustainability Safety
Low Carbon Economy					
Hamilton / Characterization - electron optics	<u>Capability:</u>	High performance electron microscopy including focussed ion beam, scanning electron and transmission electron microscopes with sub-nanometer resolution, and advanced compositional analysis using energy and wavelength dispersive spectroscopy (E/WDS), electron energy loss spectroscopy (EELS) and crystallographic texture using electron backscatter diffraction (EBSD) patterning	x	x	x
	<u>Equipment:</u>	FEI Helios NanoLab 650 dual-beam focused ion beam (FIB) microscope, FEI NanoSEM 650 field emission SEM, FEI Tecnai Osiris 200 kV field emission TEM			
	<u>Special requirements:</u>	None/available			
Hamilton / Physical simulation and processing	<u>Capability:</u>	Thermomechanical simulation of compression, tension, torsion. Also deformation dilatometry, thermal property analysis, thermal diffusivity and thermoelectric Seebeck coefficient	x	x	x
	<u>Equipment:</u>	Netzsch LFA457 thermal flash diffusivity, ULVAC ZEM-3 Seebeck, Gleeble 3800 T/M simulator, BÄHR DIL 805 quenching deformation dilatometer, BÄHR STD 811 hot torsion plastometer, Netzsch STA 449 high-temperature DSC + TGA calorimeter			
	<u>Special requirements:</u>	None/available			
Hamilton / Materials Performance	<u>Capability:</u>	Tensile, fatigue, thermo-mechanical fatigue, creep, impact, high resolution 3D real-time digital image correlation (DIC) strain mapping, crack length monitoring, and automatic video extensometry, x-ray diffractometer for retained austenite and residual stress, and nanoindentation	x	x	x
	<u>Equipment:</u>	Full suite of mechanical testing equipment up to 1000kN load, 1200C temperature and in controlled atmosphere. Drop towers to 15kJ and Charpy impact to 750J, high resolution and high speed digital image correlation (DIC),			
	<u>Special requirements:</u>	None/available			
Hamilton / Computational systems	<u>Capability:</u>	High performance computing with external access to partners	x	x	x
	<u>Equipment:</u>	XE6M Cray high performance computing facility, NVidia GPU DGX high performance work station			
	<u>Special requirements:</u>	None/available			

Hamilton - Future Relevant Equipment					
Facility/ Functional Asset	Description		MAPs	Scale-up, Demo, Standardization	Sustainability Safety
Hamilton / Materials Acceleration Platform (MAP)	<u>Capability:</u>	Robotic casting platform used to melt and cast small samples of a wide variety of materials (thermoelectrics, pyroelectrics, high entropy alloys and thermal energy storage materials) with on-board, parallel performance characterization and remote, autonomous control	x	x	
	<u>Equipment:</u>	Robot with induction casting system with in-situ XRF composition, XRD structure and other critical performance parameters such as electrical conductivity, thermal conductivity or diffusivity. Fume hood, glove box			
	<u>Special requirements:</u>	Laboratory renovation			
Hamilton / Materials Acceleration Platform (MAP)	<u>Capability:</u>	Robotic metal 3D printing platform used to additively manufacture lightweight, structurally efficient parts with on-board, parallel performance characterization and remote, autonomous control	x	x	
	<u>Equipment:</u>	Autonomous, metal 3D printing system with in-situ characterization of critical process and performance parameters. Fume hood, glove box			
	<u>Special requirements:</u>	Laboratory renovation			

Mississauga - Future Relevant Equipment					
Facility/ Functional Asset	Description		MAPs	Scale-up, Demo, Standardization	Sustainability Safety
Mississauga / Operational wet & dry labs, machine shop and mechatronics shop	<u>Capability:</u>	Synthesis of range of materials at laboratory-scale quantities, sufficient for both small-scale experiments and for development of repeatable, scalable materials production processes. Fabrication of test specimens, custom small-scale devices and systems and automated experimental apparatus.	x	x	x
	<u>Equipment:</u>	Broad range of standard chemical synthesis equipment, including ovens, reactors, mixers, spin coaters, slot-die coater, stirrers / mixers, centrifuge, ball mills, balances / scales, , lab hardware and glassware, range of standard equipment for small-scale machining and device fabrication and testing.			
	<u>Special requirements:</u>	Laboratory construction			
Mississauga / Characterization	<u>Capability:</u>	Standard and specialized state-of-the-art materials characterization capabilities capable of interrogation of broad range of optical / spectroscopic, thermo-physical, electromagnetic characteristics, at high-resolution / high-speed and in-situ. For example: parallel in-situ characterization of a broad range of chemical reactions.	x	x	x
	<u>Equipment:</u>	Various, including equipment for: optical and electron microscopy, rheometry, profilometry, surface characterization, dynamic mechanical analysis, thermal conductivity, tensiometry, optical / mass spectroscopy, thermogravimetric analysis, electromagnetic characterization and environmental simulation.			
	<u>Special requirements:</u>	Laboratory construction and flexible IT infrastructure supporting reliable, high-speed communications between laboratory equipment			

		via custom IoT devices and with remote computing infrastructure and users.			
Mississauga / Materials Acceleration Platform (MAP)	<u>Capability:</u>	Set of AI-driven robotic platforms capable of supervised and autonomous discovery of novel catalysts (including for clean generation of hydrogen and conversion of CO2 to high-value hydrocarbons) and multi-functional materials. Includes rapid on-line synthesis, parallel characterization and analysis with remote, autonomous control	x	x	x
	<u>Equipment:</u>	Set of 3 modular robotics platforms based on a common AI core, low-cost robotics hardware and adapted synthesis and characterization tools.			
	<u>Special requirements:</u>	Laboratory construction and flexible, high-speed, high-reliability IT networks capable of handling and storing large amounts of data and connecting remotely to high-performance computing systems and external system users.			

U Waterloo - Future Relevant Equipment On-site at Mississauga					
Facility/ Functional Asset	Description		MAPs	Scale-up, Demo, Standardization	Sustainability Safety
U Waterloo / Powder synthesis and characterization	<u>Capability:</u>	Synthesis, spherodization, alloying and characterization / classification of metallic and other inorganic powders for a wide range of applications.	x	x	
	<u>Equipment:</u>	Tekna plasma reactor (15kW), equipment for powder screening, classification, sizing, mixing, rheometry and density measurement.			
	<u>Special requirements:</u>	None			

U of Toronto - Future Equipment On-site at Mississauga					
Facility/ Functional Asset	Description		MAPs	Scale-up, Demo, Standardization	Sustainability Safety
U of Toronto / Processing and characterization	<u>Capability:</u>	Synthesis and characterization of catalysts for hydrogen generation and conversion of CO2 to high-value hydrocarbons.	x	x	x
	<u>Equipment:</u>	Photoreactor, operando catalyst characterization suite with mass spectroscopy, equipment for BET-chemisorption, TGA/DSC/MS, Raman spectroscopy, GISAXS / WAXS x-ray diffraction, powder XRD and environmental test chambers.			
	<u>Special requirements:</u>	None			
U of Toronto / Computational systems	<u>Capability:</u>	High-speed computing capability, supporting materials design and AI-driven MAPs.	x	x	
	<u>Equipment:</u>	HPC Cluster Expansion with 29x Lenovo Thinksystem SD530 Compute Nodes (based at SciNet)			
	<u>Special requirements:</u>	Reliable high-speed communications with SciNet			

3.6 IM/IT Strategy

The materials science research to be conducted by Mississauga / Hamilton will be data- and compute-intensive, requiring a robust IM/IT system supporting current needs while being flexible to accommodate unforeseen future needs as well as future developments in AI and Robotics. This section describes these needs based on the “SBDA Enterprise Information Management / Information Technology Needs



Narrative” (RDIMS #9898791)⁷. In addition to the research being conducted at the facility, a key success factor will be the ability to work seamlessly with other organizations which requires integration with provincial and federal networks such as ORION and CANARIE. IM/IT needs for Mississauga / Hamilton include:

Data Generation, Collection and Storage

The MAPs will be the primary generators of data within the facilities. They can potentially produce data at a rapid rate which may require local (in-lab) storage. This will be considered in the detailed design phase of those facilities and might not impact the IM/IT infrastructure of the building. For basic storage requirements, which are considered building infrastructure, the following is a good starting point:

- 240TB modules such as <https://www.backblaze.com/blog/open-source-data-storage-server/> should be considered. A full rack of these modules will provide a total capacity of 2.4PB. It would be filled incrementally based on requirements. This upper capacity limit will suffice for the foreseeable future.
- An adequately sized off-site backup solution along with the appropriate bandwidth to permit seamless data flow.
- A Laboratory Information Management System to support data acquisition, storage, and retrieval. The impact of such a system on hardware requirements will need to be determined.

Data Analysis and Visualization

A high performance cluster housed within NRC-M consisting of 8192 cores, 4 GB RAM per core, FDR InfiniBand OR Intel Omnipath, 128-256 GPU, and fast storage of 512 TB.

Some analyses and visualization tasks will require many days to conclude. During this time, data may be produced or obtained from sources both within the facility or external to it and power must not be interrupted. In addition, over time, new software and hardware coming on stream will need to be accommodated by the IM/IT infrastructure.

From Scientific Data to Decision-Making: How to Ensure the Legacy and Future of our Information

The facility will produce experimental results at a much higher rate than is typical of a non-automated one. This will increase the burden on experimental records management to enable experimental reproducibility, additional data for AI algorithms to process, intellectual property (IP) protection, and decision making. Transparency will be paramount so that future research has a solid basis to progress.

Collaboration Clusters

Collaboration is a cornerstone of the Mississauga / Hamilton initiative. They will need open and straight-forward bi-directional transfer of information from multiple sources. They will need to be well-connected and have the appropriate capacity to work with collaborators, with state-of-the-art communications tools. Some considerations are:

- Streaming video and operational data of the activities being conducted in the MAP facilities.
- Accessible and remote-controllable MAP and laboratory equipment. In the case of a MAP consideration needs to be given to the type of user interaction. For example, safety has to be paramount - one can't let a remote user take control of robotic movement; however asking for a different chemical ratio along with a measurement of the result could be acceptable.
- Capability for instruments to be connected so staff can access them from outside a facility (perhaps through VPN), with some accessible by vendors for troubleshooting and upgrading.
- Users must be able to login remotely, from their own device (i.e. NOT just GoC issued equipment) using a protocol such as Secure Socket Shell (SSH). For authentication, consider Google's Two-Factor Authentication (2FA).

⁷ This report was spearheaded by the Canadian Food Inspection Agency and does an excellent job of describing the IM/IT issues faced by research organizations.



- Lab segregation is important: labs need to be linked to one another, and not connected to other corporate or secure systems.
- Video conferencing equipment should be ubiquitous, easy to use and work with multiple standards: Skype, Google hangouts, etc. In some cases, having an “always on” video link in the coffee/quiet rooms of an ongoing, geographically dispersed, partnership will be needed. Systems that support microphone arrays that automatically isolate the speaker (via video) in a room full of people should be supported, as well as systems that allow latency-free telepresence and simultaneous interaction and visualization of 3D objects.
- Collaboration software such as Google docs, Dropbox, and Slack will need to be permitted as will flexibility for future developments.

Connectivity

The IM/IT infrastructure must enable seamless access to external partners and geographically-dispersed NRC and NRCan staff. Things to consider include:

- Wi-Fi: Ideally 802.11ac, but what matters more is that it needs to be something simple that off-the-shelf webcams and components can connect to. Certificates and excessive authentication won't work with simple IoT sensors. Wi-Fi needs to be able to support users inside and outside a firewall.
- Internal network: Must be wired for 10 GB network cards, with cabling for 100-400GBs for future upgrades. Properly sized conduits are required to allow upgrades to future protocols.
- 100 GBps switches allowing full-speed connection to ORION (<https://www.orion.on.ca/about-us/our-network/>).
- Fibre runs (at minimum necessary conduit) from the cabinet room to the MAP labs, providing an option for future ultra-high speed connections.
- Bandwidth into and out of the building must be at least 100GBps.

Business Continuity and System Flexibility for Scaling-Up and Surge Capacity

It is reasonable to expect that Mississauga / Hamilton will be conducting research on materials of national importance. The operation of certain facilities within each site may become critical for a period of time. Mission-critical systems must remain operational under circumstances such as natural disasters, power failures, and unpredicted surges. Arrangements must be put in place with SSC to rapidly repair or procure critical equipment/infrastructure in the event of an unforeseen upset.

Infrastructure must be designed to accommodate increases in computational and network capacity with an on-going life-cycle planning and management approach to not only ensure a solid IM/IT infrastructure but compatibility with outside partners.

3.7 Security

As the focal point of a national network of a diverse range of collaborating partners with facilities accessible both on-site and remotely, Mississauga / Hamilton will need to give very careful consideration to both physical and IM/IT security issues. This will be critical to ensure both efficient access to facilities and data to authorized individuals and effective denial of access to unauthorized parties. Mississauga / Hamilton will conduct work with a range of levels of sensitivity, from unrestricted and fully-open / accessible, to confidential / business-sensitive and even ITAR-restricted and Secret-level classified.

3.7.1 Physical Security

General access to the Mississauga / Hamilton facility will be badge-restricted with further badge-access permissions required for entry to *all* office areas and laboratory work spaces and some individual laboratories and office / meeting spaces with an additional level of access-control. At least one mid-sized meeting room for approximately 10-15 people will be designed for Secret-level discussions (Sensitive Compartmented Information Facility (SCIF)).



3.7.2 IM/IT Security

IT security will be designed on a similar basis as physical security with secure Wi-Fi available to all on-site visitors and controlled access to various information networks based on individual need. The sites will have the ability to provide wired connections to specific locations such as collaborator / tenant laboratories that connect to the internet, but not to government networks. Remote access to the facilities and their data networks, which will be critical to make effective use of the MAPs and other platforms, will be available both to government staff and others as appropriate. Consideration will also have to be given how to address issues with IoT devices, which are certain to play an important role in the MAPs and other experimental apparatus. These and other considerations will require network flexibility to, for example, set temporary isolated networks.

3.8 Site Considerations

3.8.1 Type of Science

CMAT and NRC-M are located just 40 km apart, offering the potential for agile realization of excellence in science. Nestled within Canada's manufacturing heartland and centred among 7 world-class universities, Mississauga / Hamilton is ideally situated to capitalize on the close proximity to key industry and academic stakeholders in advanced and clean energy materials. This close proximity is a key enabler of collaboration and scientific innovation, facilitating fluid mobility of staff among sites, co-location of partner researchers and access to each other's state-of-the art facilities. Proximity has helped to fuel existing partnerships with XRCC (directly adjacent to NRC-M), University of Toronto (campuses 5 km and 30 km from NRC-M) and McMaster University (2 km from CMAT). Nearby Toronto and Hamilton airports render Mississauga / Hamilton easily accessible to researchers from across the country and around the world.

3.8.2 Alignment with Laboratories Canada Guiding Principles

- Scientific Innovation: a national network that will radically accelerate the discovery, development, integration and demonstration of disruptive new materials that will drastically reduce GHG emissions
- Collaboration: at the core of the proposed national network, enabled by co-location of NRCan, NRC and partner staff, fluid mobility and access among facilities and coordinated strategic research directions
- Functionality and Modernization: achieved through the new, state-of-the art facilities tailored to the scientific program needs
- Environmental Sustainability: achieved through the scientific program (discovery of disruptive new clean energy materials, more rapid and efficient scaling and development of new materials) and the use of green building technologies and practices
- Universal Accessibility: achieved by offering physical access to the Mississauga / Hamilton facility, for example co-location or other forms of collaboration, as well as virtual access to data and MAPs
- Optimization of Economic and Enterprise Value: coordinated effort to integrate, leverage and build upon existing national infrastructure, capabilities and initiatives at NRC, NRCan and partners

4 Risks

As shown in Table 4-1, most of the risks associated with establishment and operation of the proposed initiative are similar to those envisioned for other Laboratories Canada initiatives, such as those associated with securing building capital and effectively managing site(s) construction. Unusually high risks in this case relate to the criticality of high-performance IT / IM (assessed as Very High risk) and securing remaining needed capital funding for acquisition of equipment (assessed as High risk) (see Table 4-2).



Table 4-1. Project Risks

No.	Risk Rating	Risk Description	Planned Mitigation
R1	High	Insufficient operating funds to support capacity utilization	1. Incremental joint resource allocation at CMAT, partially transferring strategic assets to NRC-M 2. Develop strong academic and industrial partnerships that deploy human and physical assets in advance of full resource deployment
R2	High	Insufficient future capacity to address demand and facility flexibility	1. Prepare facilities for future expansion during facility construction 2. Identify partner assets to transfer or expand capacity
R3	High	Project delays due to approval timing and governance uncertainty	1. Terra Canada program team establishes clear and pragmatic governance and approval processes 2. Terra coordinates approvals within the governance structure
R4	Low	Missed critical supporting science requirements	1. NRC, NRCan and nearby academic institutions have existing high calibre scientific capabilities with Hamilton offering flexibility with manpower and equipment capacity
R5	High	Sufficient budget for construction	1. NRC and NRCan to scale as possible within the budget allocation while compromising the science program
R6	Very High	Integration of IM/IT user requirements for IM/IT	1. Close coordination and cooperation between NRC/NRCan and SSC 2. Continued insistence that SSC focus on delivery of requirements versus focus on policy
R7	Medium to High	Interruption of client's operations	1. Minimal impact at CMAT due to small scale of renovation and ease of segregating work 2. Moderate-high impact to NRC-M first floor operations during 3 rd floor construction can be minimized through close coordination between research operations and construction plans
R8	High	Insufficient capital for equipment	1. Advanced business planning

Table 4-2. Risk Matrix

		Consequences				
		Negligible	Minor	Moderate	Major	Severe
Likelihood	A: Almost Certain				R6	
	B: Likely			R5	R1, R3	
	C: Possible		R4	R2, R7, R8		
	D: Unlikely					
	E: Rare					
	Low		Medium	High		Very High



5 Source of Funds

Funding for 2,405 m2 was approved via the May 2019 TB submission and approval. An increase in space and budget will need to go to LC governance for approval. Laboratories Canada to act as project sponsor.

6 Approvals

Laboratories Canada has indicated that the governance approval process for SSORs is at the determination of each cluster – at a minimum Laboratories Canada requires that SSORs go through DGCC governance approvals.

6.1 Approval of this Science Statement of Requirements and to Initiate Further Documentation

Terra DGCC

Representative	Department	DATE	SIGNATURE
Frank Jefferies Director General of Administrative Services and Property Management	NRC		
Terry Hatton, Director General of Real Property & Workplace Services	NRCan		
Joanna Ankersmit Director General Program Implementation	Laboratories Canada		



APPENDIX A Space Typology Definitions

Dry Laboratory: Scientists work with dry stored materials, electronics, and/or large instruments with few piped services. It may require temperature, humidity and dust control and clean power.

Wet Laboratory: Scientists work with chemicals, drugs, or other material or biological matter, where they are tested and analyzed requiring water, direct ventilation, and specialized piped utilities.

Containment Level: A containment level is a set of containment precautions required to isolate dangerous agents in an enclosed laboratory facility. The levels of containment range from the lowest containment level 1 (CL-1) to the highest at level 4 (CL-4).

Containment Level 1 (CL1): This does not require special design features beyond those suitable for a well-designed and functional laboratory. Containment safety cabinets are not required. Work may be done on an open bench top, and containment is achieved through the use of practices normally employed in a basic microbiology laboratory.

Containment level 2 (CL2): The primary exposure hazards associated with agents requiring CL2 are through physical contact. Agents requiring CL2 facilities are not generally transmitted by airborne routes, but care must be taken to avoid the generation of aerosols or splashes. It is sufficiently sealed to maintain the required pressure differential (negative or positive) between adjacent rooms or suites. The HVAC system is only required to provide sufficient air changes per hour. Primary containment devices such as a centrifuges with sealed rotors or safety cups are to be used as well as appropriate personal protective equipment (i.e., gloves, laboratory coats, protective eyewear). As well, environmental contamination must be minimized by the use of handwashing sinks and decontamination facilities (autoclaves).

Containment Level 3 (CL3): Scientists handle agents that may be transmitted by airborne route, often in a low infectious dose and that can cause serious or life-threatening disease. CL3 emphasizes additional primary and secondary barriers to minimize the release of infectious agents into the immediate laboratory and the environment. All joints, connections and penetrations, including doors and windows, electrical outlets, lights, etc., are to be fully sealed to ensure no air leakage from lab, other than through the dedicated exhaust system. Additional features to prevent transmission of CL3 agents are appropriate respiratory protection, HEPA filtration of exhausted laboratory air that is independent of other areas and strictly controlled laboratory access.

Base Isolation: The facility is built to minimize the vibration that will limit the operational tolerance or the detection limits of scientific equipment. This is usually achieved by providing a partial separation of the building structure from potential vibrations generated by vehicle traffic, trains, earthquake motion, building mechanical operations or human circulation by preventing the building superstructures from absorbing the vibration energy.

Radiation containment: This containment represents an isolated space with thick concrete walls and in some cases completely independent structures. In order to reduce airborne releases, highly efficient filters and radiation monitors must be installed as part of the ventilation systems.

Engineering Bay/Warehouse space: These facilities provide a proper environment for the purpose of researching with the use of bigger equipment that won't fit in a regular building and that require protection from the elements. Engineering Bay/Warehouses spaces must be designed to accommodate the loads of the equipment, the associated handling of the equipment, and the needs of the operating personnel. These spaces are generally larger, have higher ceilings and space clear of structure, such as hangar facilities or a workshop.

Lab related Offices: Enclosed office located adjacent to the laboratory room with easy access for lab user.



Auditorium/ Bd. Rm/ Training Rm: These are areas for large meetings, presentations, and performances. Auditorium space type facilities may include assembly halls, exhibit halls, auditoriums, and theaters. Auditorium space types do not include such features as sound reinforcement systems, audiovisual systems and projection screens, food service facilities, proscenium stages with heights greater than 50'–0" or fly gallery, orchestra pits, revolving or hydraulic stage platforms, flying balconies, movable seating, or billboard systems.

Archive/Records/Library: These are areas where bound paper documents, film, or magnetic media are stored. An Archive space type may include both open and closed storage systems and moveable shelving systems, and be applicable to file rooms and other dense storage of material in conditioned office environments.

Hazardous Storage: These are spaces used for storage of chemical or hazardous materials. Safe storage of chemicals must begin with identification of the chemicals to be stored and their intrinsic hazardous properties. Since many chemicals have several hazards, which may vary in degree of severity, depending on quantity and concentration, spaces may require temperature, ignition control, ventilation, segregation and identification.

Light Industrial/General Storage: Spaces are used for the assembly, disassembly, fabricating, finishing, manufacturing, packaging, and repairing or processing of materials. Light Industrial space types can include but are not limited to spaces for printing, commercial laundry, photographic film processing, vehicle repair garages, building maintenance shops, metal work, millwork, and cabinetry work. Includes a shipping and receiving dock for trucks and vans, a staging area, and at least one office space for dock supervisors or managers.

Lobby/Core/Circulation: The lobby Includes foyers, entries to halls, and security screening areas at or near the entrance to a building or demarcated space, and are meant to welcome and direct tenants and visitors, control access, and provide exit ways from buildings. This space type is often designed with both secure and non-secure areas. The core includes the central area of a building housing the elevator and stairwells, electrical equipment, restrooms, other facilities and all the circulation area.

Food Services: Food Services includes cafeterias, sandwich shops, coffee shops, fast food retail, and other food services that involve the preparation and handling of food items for the consumer. Food Service space types are distinguished from other spaces where food may be vended (such as employee lounges) by the health and sanitation requirements related to the handling of unpackaged food products and/or processing of non-disposable dishware.

Server/ Computer room: Is a facility for "vertical" data processing equipment including dense frame and rack-mounted processing systems with critical uptime requirements.

Parking: Surface: This refers to large paved areas used for extensive vehicle parking—beyond the incidental parking provided for individuals, official government parking, and short-term drop off—located adjacent to a building.

Parking: Outside/Structured: This refers to an above-grade, ramp access, open-air structure specifically designed to accommodate vehicle parking.



APPENDIX B Science Plan

The following text excerpts relating to Mississauga / Hamilton are taken **from Terra Canada Science and Innovation Cluster Science Plan, Draft 05 April 2019:**

Areas of Science (Page 4)

Low Carbon Economy: Supporting Canada's transformation to a low carbon economy within the following horizontal platforms:

- *Low Emission Transportation:* Technology development for carbon neutral through to zero emissions vehicles across transportation modes (air, land, water).
- *Buildings and Infrastructure:* Technology development to reduce carbon in buildings and infrastructure, as well as reduce energy consumption, construction and operating costs, and residual impacts on people and the environment.
- *Advanced Materials and Integrated Energy Systems:* Systems integration to accelerate discovery and development of clean energy materials, from laboratory to application.

Locations (Page 4-5)

Mississauga/Hamilton: This region will integrate the NRC Innovation Centre in Mississauga, with the NRCan CanmetMaterials in Hamilton, as a first-of-its-kind facility in Canada focused on **Low Carbon**

Energy research to accelerate discovery and development of advanced materials, particularly in the clean energy domain.

This theme will enable a first-of-its-kind Canadian network, with researchers in the **NCA working closely with** Terra Canada regional locations in **Mississauga and Hamilton** (competencies in Accelerated Materials Discovery, Production Scale-up; Standardization; and Materials and Process Sustainability and Safety)

Low Carbon Economy (Page 8)

Significant, critical mass effort will be required from scientists across Canada to help the Government meet ambitious priorities and targets outlined in the Clean Technology Economic Strategy Table, the Paris Accord, the Pan-Canadian Framework on Clean Growth and Climate Change, and Mission Innovation.

Terra Canada will help Canada meet emissions reduction targets, and grow the clean energy and manufacturing sectors. Research will focus on reducing the carbon footprint from hydrocarbon-based fuels and fuel consumption (e.g., vehicles, buildings), with an ultimate goal of replacing the need for these fuels through electrification. It will integrate multidisciplinary expertise ranging from advanced materials to batteries to alternate fuels/biofuels. Terra Canada will also enable testing of innovations under various real-world and laboratory conditions – engine/vehicle testing, certification of mining engines (linking to the Sustainable Land and Resource Development theme), energy management analysis, and test buildings and infrastructure. Some early examples would be to expand integrated research on GHGs, NOx, particulate matter, and related constituents in the atmosphere (transport, chemical modification, secondary aerosol formation, removal); and ensuring the human health and air quality benefits of clean technologies.

This theme will enable a first-of-its-kind Canadian network, with researchers in the **NCA working closely with** Terra Canada regional locations in **Mississauga and Hamilton** (competencies in Accelerated Materials Discovery, Production Scale-up; Standardization; and Materials and Process Sustainability and Safety) and **Winnipeg** (competencies in Sustainable Materials Manufacturing, and Additive and Digital Manufacturing). The network will expand beyond Terra Canada labs – seeking to establish knowledge sharing with air quality/climate modelling R&D in Montreal/Victoria, and advanced materials and



manufacturing R&D in Boucherville and Montreal, and energy R&D in Vancouver, Devon, and Varennes, as well as ECCC's air quality/climate modelling in Montreal/Victoria.

Research will be conducted within three horizontal platforms: (1) Low Emissions Transportation, (2) Buildings and Infrastructure, and (3) Advanced Materials and Integrated Energy Systems

Bringing together expertise and facilities in areas such as: Measurement science (black carbon metrology), Battery testing, Biogas and electrochemistry, Hydrodynamics, Fuel storage and test engines, Diesel research and engine certification, Emissions research and measurement, Construction and housing technologies (roofing systems, insulation teams, and facades), and Advanced materials and additive manufacturing, along with emerging technologies (nanochemistry, nanoproduction, printable electronics)

Table1: Science and Research Platforms of Terra Canada (Page 13 – 14)

Platform 6: Advanced Materials and Integrated Energy Systems

Terra Canada partners will accelerate advanced materials development for integrated energy systems. The innovation process for the development of new materials, from laboratory to application, is notoriously expensive and slow often requiring 20 years to realize. Canada has strong academic and government capabilities in materials science, however there is a considerable opportunity to accelerate the discovery and development of clean energy materials and lead in the commercialization of advanced materials.

Terra Canada proposes to create a national network to develop innovative materials, scale production, and demonstrate applications, by accelerating the discovery of new clean energy materials by a factor of >10. Through close collaboration with Mission Innovation Clean Energy Materials Innovation Challenge (IC6), which Canada co-leads, Terra Canada will aim to accelerate the discovery of new materials for clean energy systems through artificial intelligence-driven, computing-enabled, robotic platforms – called the Materials Acceleration Platform (MAP). The MAP would be configured to develop organic, inorganic, and functional materials such as thermoelectric generators, which transform waste heat into electricity, or new high entropy alloys that have unique properties such as extreme strength and corrosion and wear resistance, which are important for most clean technology applications.

Multifunctional materials development and characterization will take place for energy production, conversion and storage, including standardization, safety, and sustainability (recyclable/biodegradable) in a whole lifecycle approach. Materials researchers will also collaborate with the additive manufacturing sector to enable novel structures with a lower carbon footprint. For example, multifunctional materials for polymer additive manufacturing (3D printing) will be developed, creating a library of 3D printable smart materials to enable direct manufacturing of structural components and end-products that incorporate functional responses (e.g., strain sensing, self-healing, shape-change and energy harvesting).

New materials will be de-risked through Terra Canada to aid technology transfer to the private sector.

Collaborations and Engagement (Page 17)

The **National Research Council of Canada** is establishing a new collaboration centre with University of Toronto to work together in Green Energy Materials. Focus will be on developing and demonstrating emerging green energy materials and associated production processes (next generation photovoltaics, renewable carbon-based feedstocks, renewable fuels and fuel cells next-generation, consumer-scale energy solutions). Efforts will include technologies for conversion of CO₂ to hydrocarbon fuels.

Science Plan Appendix 1: Terra Canada Regional Locations (Page 26)

Mississauga/Hamilton

This Terra Canada satellite will integrate the NRC Innovation Centre in Mississauga (an addition to a 3400m² NRC facility, now under construction), with the NRCan CanmetMaterials in Hamilton (renovation



of existing 100m² laboratory), as a first-of-its-kind network in Canada focused on accelerating discovery and development of a wide variety of advanced materials, including those in the clean energy domain. Staff from NRCan and NRC will work in both facilities, developing strong synergies and cross-functional training to build unique capabilities in Accelerated Materials Discovery and Process Development; Production Scale-up Demonstration and Standardization; and Materials and Process Sustainability and Safety. This satellite will work closely with Terra Canada NCA and Terra Canada Winnipeg to deliver outcomes under the **Low Carbon Energy** theme.

Partnerships and collaborations within the region will be explored, initially with universities. Efforts will extend or build upon existing engagements, such as:

- A new collaboration has recently been established with the University of Waterloo with the Multi-Scale Additive Manufacturing lab to develop precursor materials for additive manufacturing. Research will centre on understanding the influence of variations in properties of materials feedstock for additive manufacturing (AM), to lower the cost of material feedstock and develop new material feedstock, thereby realizing uniformly high quality, novel, and lower cost AM-derived products. This collaboration will establish technical competencies in metal powder based synthesis and characterization, develop the understanding of the metal powder feedstock used in AM to create an industrially robust process, as well as lower the economic barriers to metal AM by increasing the accessibility of low-cost materials and processes.
- Located adjacent to the NRC Mississauga facility and accessible through collaboration with NRC, Xerox Research Centre of Canada's (XRCC) materials design, analysis, scale-up and pilot plant facilities are critical enablers cost-effectively scale the production of new material discoveries from laboratory quantities to volumes needed to advance past the proof-of-concept stage. Moreover, NRC and XRCC are founding partners of the Canadian Campus for Advanced Materials Manufacturing (CCAMM), a flagship hub for sharing synergies and facilities from industry, public, and academic investments to de-risk the development and deployment of commercially viable materials technologies and bridge the gap between laboratory demonstration and industrial scale manufacturing.
- NRCan's long-standing collaborations with **McMaster University** in the development of advanced materials, advanced imaging, and material characterization, offer a model for on-going NRC discussions with McMaster in the same subject areas.
- NRCan's on-going partnership with the **University of British Columbia** and the **University of Toronto** in the area of AI, advanced robotics, and materials for clean energy applications, in which NRC also aims to become a core partner.
- NRC has signed an agreement with **University of Toronto** to set up a Collaboration Centre to work on an advanced materials challenge program that contribute to IC6 of the Mission Innovation Initiative.
- NRCan and NRC have a mutual focus on materials innovation and support for advanced manufacturing. Respective research groups (**NRCan-CMAT** and **NRC in Saguenay** (Aluminium Technology Centre) and **Boucherville** (Industrial Materials) work together to maximize impact on industry and society, currently in the area of materials for lightweight, fuel-efficient vehicles.

Science Plan Appendix 2: Terra Canada Vignettes: **ADVANCED MATERIALS FOR CLEAN ENERGY**

Terra Canada federal organizations involved: NRC, NRCan

ISSUE: Advanced materials underpin emerging and disruptive clean energy technologies but are costly, time-consuming to develop and risky to bring to market, limiting their full exploitation and integration into products, across industry from small-medium enterprises to multinationals.

OUTCOME: Canada de-risks technology adoption for Canadian industry driving accelerated discovery, integration and demonstration of advanced materials innovations to drastically reduce GHG emissions.

Current State:



Under the Paris Agreement, Canada committed to reducing net GHG emissions by 30% below 2005 levels by 2030. The Pan-Canadian Framework on Clean Growth and Climate is a plan designed to grow the Canadian economy while reducing emissions and building resilience to adapt to a changing climate.

A transition to a new environmentally friendly energy future, through drastic GHG emission reductions, is essential for Canada to achieve its ambitious climate change goals. Economically competitive, environmentally friendly and spill-safe energy alternatives to fossil fuels must be found. New materials, such as catalysts or solar cell components, are critical enabling blocks in the pathway to clean energy. The innovation process for the development of new materials from laboratory to application is notoriously expensive and slow, often requiring 20 years to realize. Canada has strong academic and government materials science capabilities, but there is considerable opportunity to accelerate the discovery and development of clean energy materials and lead in their transition to market.

What Terra Canada will bring to Canadians:

Building and leveraging upon recent investments and partnerships, Terra Canada would integrate and expand on Canada's world-class materials R&D capabilities to establish a first-of-its-kind Canadian network able to accelerate discovery and development of novel disruptive clean energy materials.

It will establish and operate a national innovation network supporting foundational R&D to develop new materials, scale their production, and de-risk and demonstrate application.

Strategically coordinated facilities of NRC and NRCan will form a national network to integrate and exploit national R&D investments, competencies and initiatives in clean energy materials discovery, notably:

- In accelerating advanced materials technology development and commercialization. The emphasis is on platform technologies and competencies applicable to multiple material families with a focus on materials for clean energy and advanced manufacturing. Examples include photo- electrochemical conversion of CO₂ into building blocks for fuels; materials development for primary functional process requirements (e.g., catalyst for reduction of CO₂); and novel feedstock materials for 3D printing.
- In accelerating materials discovery and development. Through close collaboration with Mission Innovation Clean Energy Materials Innovation Challenge (IC6), which Canada co- leads, Terra Canada will aim at accelerating discovery of new clean energy materials by a factor of >10 through artificial intelligence-driven, computing-enabled, robotic platforms – called the Materials Acceleration Platform (MAP). The MAP would be configured to develop organic, inorganic and functional materials such as thermoelectric generators, which transform waste heat into electricity, or new high entropy alloys that have unique properties such as extreme strength and corrosion and wear resistance, which are important for most clean technology applications.

Terra Canada will accelerate and de-risk adoption of promising new energy materials technologies, link new materials to applications, and facilitate technology transfer applications reduce GHG.

Key Partners and Stakeholders:

- Academic partners, e.g. University of Toronto, University of British Columbia, University of Waterloo, McMaster University, University of Alberta, INRS, Université de Montreal.
- NGOs, including NGM Canada (Advanced Manufacturing Supercluster), Green Centre Canada
- Industry partners and industry associations, such as Xerox Research Centre of Canada, BASF, Nanoexplore, Canadian Manufacturers and Exporters

Example of equipment or facility efficiency gains:

Combining under Terra Canada multiple requirements for equipment and facilities for a broad range of materials synthesis, characterization, testing and production scale-up will allow optimization of use of such highly-specialized facilities and support improved efficiencies in collaborative activities. For example, the Terra Canada network could scale-up and improve the MAP, which is utilized to deliver the IC6 goals.



APPENDIX C Hamilton Business Case

TERRA CANADA INVESTMENT IN NRCAN-HAMILTON

Introduction

Terra Canada's 'Low Carbon Economy' theme helps Canada meet emissions reduction targets, grow clean energy and manufacturing sectors, and advance development of clean technologies. A key pillar in tackling these challenges is access to novel materials key to clean energy solutions. However, it can take decades to commercialize new materials, often making it impractical / uneconomical for businesses and investors.

Terra Canada will establish via NRC-Mississauga / NRCan-Hamilton a national network to develop and deploy a set of integrated technology platforms to accelerate the discovery of new materials and processes that will underpin the commercialization of disruptive new products in support of Canada's low-carbon future. Central to these efforts is a revolutionary approach to materials discovery, Materials Acceleration Platforms⁸, MAPs: open-access, artificial intelligence (AI)-driven, smart robotics platforms for autonomous materials development.

NRCan funded a first-of-a-kind MAP demonstration project "Ada"⁹ at the University of British Columbia in collaboration with University of Toronto that focusses on functional materials for clean energy innovation (improved organic solar cells, catalysts for CO₂ conversion). NRC-Miss and NRCan-Ham are proposing the creation of permanent MAPs installations accessible to industry and academe in an existing, highly collaborative research environment.

NRCan-Ham's MAPs would leverage its unique capabilities on structural (3D printed metals) and specific functional materials and NRC-Miss' MAPs would focus on functional materials (CO₂ conversion and H₂ production). Each MAP platform focusses on a specific class of materials and processes, however extensive learning and co-development using open data structures will exponentially accelerate materials discovery and optimization. Through the dedicated MAPs facilities, Canada will develop a leadership role in clean technology materials innovation and, therefore climate change mitigation.

This proposal aligns with Terra Canada's goals of collaborative research in modern, shared facilities that break through current silos that hinders federal S&T. Enabling this kind of collaborative R&D proposed by NRCan and NRC is already attracting global attention and partners, better positioning federal S&T to bring forth innovation with a diverse and inclusive talent pool. Most importantly, NRCan provides for an agile and responsive foray in delivering science excellence through a revolutionary platform approach.

Rationale for Laboratories Canada Investment in NRCan-Hamilton

NRCan-Ham focuses on structural (e.g. metals and composites) and functional materials testing and processing for clean energy applications across energy end-use, distribution and production. The lab has unique, pilot-scale facilities for materials casting, metal forming, enabling technologies (e.g. joining), and sophisticated characterization.

NRCan-Ham is using the Terra Canada initiative to operationalize three cutting-edge technologies: high performance computing, smart robotics and AI to accelerate materials development. The lab is capable of **commissioning a MAPs facility within one-year of approval**, but requires a modest investment in infrastructure updates. Capital equipment procurement and staffing plan reviews have been initiated to

⁸ Materials Acceleration Platform report, <http://mission-innovation.net/wp-content/uploads/2018/01/Mission-Innovation-IC6-Report-Materials-Acceleration-Platform-Jan-2018.pdf>

⁹ Project Ada, <http://www.projectada.ca/>



bring critical new expertise to support the transition to the high technology MAPs approach.

Under Terra Canada, NRCan-CMAT and NRC-Miss are requesting capital funds that will **establish first-of-its-kind facilities in Canada** but expand and leverage state-of-the-art facilities in a cost effective and timely manner, a clear demonstration of responsible public stewardship. Staff from NRCan and NRC will co-locate at both facilities, first in NRCan-Ham then in NRC-Miss, developing strong synergies. Both facilities will rely on the expertise and unique equipment currently available at NRCan-Ham and those to be available at NRC-Miss.

The NRCan-Ham facility specifically aims to deploy two MAPs, the first of which has largely received operational and capital funding:

1. Thermoelectric materials and potentially high entropy alloys ('HEA') using casting technology
2. Materials for additive manufacturing such as powders and metal 3D printing.

NRCan-CMAT is requesting modifications to its existing building, **representing an early win for Laboratories Canada**, developing knowledge that would **de-risk the NRC-Miss facility expansion**, and establishing early science partnerships between the two departments and integration of the new platforms.

This investment would **use funds allocated under the Terra Canada Hamilton-Mississauga project**. An accelerated schedule is possible as the space already exists and will only require enhancements to deploy MAPs. Specifically, NRCan-Ham requires a simple laboratory fit-up involving renovation of existing space (91.5 m²), including provisions for high-speed data connection (CANARIE), improved lighting, ventilation and humidity controls and electrical work. This aligns with Terra Canada's goals of building living labs with high-speed connections to facilitate science excellence and would be integrated in a LEED Platinum building less than ten years old.

An architectural study was recently completed and **the total cost to fit-up NRCan-HAM is \$1.37M (Class "C" estimate and landlord build costs)**. Discussions between McMaster Innovation Park ("MIP", the building owner) and NRCan confirmed that building modifications are permissible under NRCan's lease agreement with MIP and could be completed in eight months. The 25+25+20 year lease options make it a highly favourable, cost effective approach. In this case, Laboratories Canada will play a coordinating role (with DCAC) but will not necessarily require A&E or CM contracts.

Laboratories Canada: Enabling Science Through Real Property

An investment in NRCan-CanmetMATERIALS to deploy MAPs would catalyze impacts across the materials R&D and manufacturing landscape in Canada:

- *Meeting Canada's Clean Growth Goals.* This proposal links to Terra Canada's Low Carbon Economy theme by accelerating the discovery and development of materials that will help Canada meet emissions reduction targets, and grow the clean energy and manufacturing sectors, with notable advancements in clean technology.
- *Doing Science Differently.* The proposal will enable the open collaboration and integration targeted by Laboratories Canada, promoting scientific excellence by growing collaboration with world-class researchers. Notably, NRCan and NRC scientists will work with researchers from the Vector Institute for Artificial Intelligence, the Canadian Institute for Advanced Research, University of British Columbia and University of Toronto. Together with these partners, and by leveraging direct linkages to the advanced manufacturing supercluster, NGen, this new NRC-NRCan partnership will become a hub and a catalyst for advanced materials innovation and commercialization in Canada.
- *Canadian Clean Technology Leadership.* As the first such facility in the world, the Terra Canada Advanced Materials Centres in Hamilton and Mississauga will position Canada as a leader in the emerging and transformative field of accelerated materials discovery using AI and smart robotic.